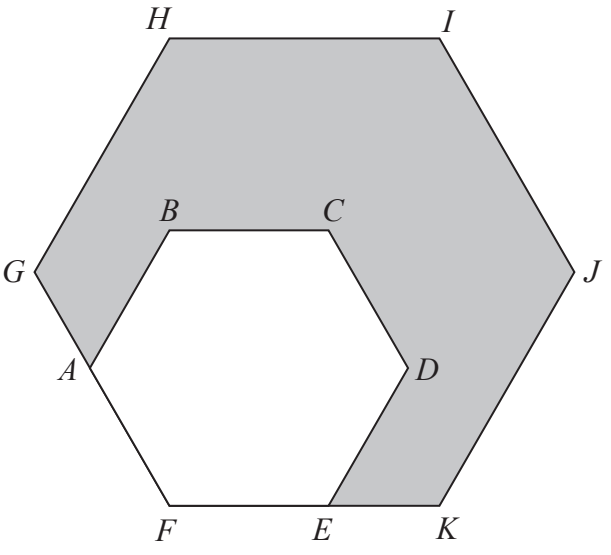
19



ABCDEF is a regular hexagon with sides of length x.
This hexagon is enlarged, centre F, by scale factor p to give hexagon *FGHIJK*.

Show that the area of the shaded region in the diagram is given by $\frac{3\sqrt{3}}{2}(p^2 - 1)x^2$

A regular hexagon comprises 6 equilateral triangles.

One triangle of side x has area ½x²sin60° = √3x²/4.

Inner hexagon: 6 × √3x²/4 = 3√3x²/2.

Enlargement p multiplies area by p².

Shaded area = 3√3p²x²/2 – 3√3x²/2

= (3√3/2)(p² – 1)x², as required.

4 marks total
Triangle area
Inner hexagon
Outer p² area
Subtract to result

(Total for Question 19 is 4 marks)

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